

PROBLEMS 21.3

1. Find $L \left(\int_0^t e^{-t} \cos t \, dt \right)$.

2. Given $L \{2\sqrt{(t/\pi)}\} = 1/s^{3/2}$, show that $L \{1/\sqrt{(\pi t)}\} = 1/\sqrt{s}$.

(U.P.T.U., 2005 ; Madras, 2003)

3. Given $L [\sin (\sqrt{t})] = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4\pi}$, prove that $L \left[\frac{\cos (\sqrt{t})}{\sqrt{t}} \right] = \sqrt{\left(\frac{\pi}{s} \right)} e^{-1/4\pi}$.

(Mumbai, 2009)

Find the Laplace transforms of the following functions :

4. $t \sin^2 t$ (Nagarjuna, 2008)

5. $\sin 2t - 2t \cos 2t$.

(Anna, 2003)

6. $t^2 \cos at$.

7. $t \sinh at$.

8. $te^{2t} \sin 3t$. (Madras, 2003)

9. $te^{-2t} \sin 4t$.

(V.T.U., 2008)

10. $t^2 e^{-3t} \sin 2t$. (Madras, 2000 S)

11. $(e^{-at} - e^{-bt})/t$.

(Anna, 2005 S)

12. $(\sin t)/t$. (P.T.U., 2010)

13. $\frac{(\sin t \sin 5t)}{t}$.

(Mumbai, 2008)

14. $(e^{at} - \cos bt)/t$. (U.P.T.U., 2003)

15. $(e^{-t} \sin t)/t$.

(V.T.U., 2009 S)

16. $(1 - \cos 3t)/t$. (V.T.U., 2006)

17. $(1 - \cos t)/t^2$.

(Hazaribag, 2008)

18. $2^t + \frac{\cos 2t - \cos 3t}{t} + t \sin t$.

(V.T.U., 2004)

19. Evaluate (i) $\int_0^\infty \frac{\cos 6t - \cos 4t}{t} dt$

(Mumbai, 2008 ; P.T.U., 2006)

(ii) $\int_0^\infty \frac{e^{-\sqrt{2}t} \sinh t \sin t}{t} dt$ (Mumbai, 2005)

(iii) $\int_0^\infty te^{-2t} \sin 3t \, dt$

(V.T.U., 2008)

(iv) $\int_0^\infty te^{-t} \sin^4 t \, dt$.

20. Prove that (i) $\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt = \log \frac{b}{a}$.

(S.V.T.U., 2009 ; Mumbai, 2007 ; J.N.T.U., 2006)

(ii) $\int_0^\infty \frac{e^{-2t} \sinh t}{t} dt = \frac{1}{2} \log 3$ (Mumbai, 2008)

(iii) $\int_0^\infty \frac{e^{-t} \sin t}{t} dt = \frac{\pi}{4}$.

(V.T.U., 2009 S)

(iv) $\int_0^\infty \frac{e^{-t} \sin^2 t}{t} dt = \frac{1}{4} \log 5$.

(Kurukshetra, 2006)

21. Evaluate (i) $L \left(\int_0^t \frac{\sin t}{t} dt \right)$

(J.N.T.U., 2005)

(ii) $L \left(\int_0^t e^{-t} \cos t \, dt \right)$

(iii) $L \int_0^t \frac{e^t \sin t}{t} dt$.

(P.T.U., 2009 S ; S.V.T.U., 2009 ; Bhopal, 2008)

22. Show that (i) $L [t J_0(at)] = \frac{s}{(s^2 + a^2)^{3/2}}$ (ii) $\int_0^\infty te^{-3t} J_0(4t) \, dt = 3/125$.

21.12 INVERSE TRANSFORMS — METHOD OF PARTIAL FRACTIONS

Having found the Laplace transforms of a few functions, let us now determine the inverse transforms of given functions of s . We have seen that $L \{f(t)\}$ in each case, is a rational algebraic function. Hence to find the inverse transforms, we first express the given function of s into partial fractions which will, then, be recognizable as one of the following standard forms :

(1) $L^{-1} \left[\frac{1}{s} \right] = 1$.

(2) $L^{-1} \left[\frac{1}{s-a} \right] = e^{at}$.

$$(3) L^{-1} \left[\frac{1}{s^n} \right] = \frac{t^{n-1}}{(n-1)!}, n = 1, 2, 3, \dots$$

$$(4) L^{-1} \left[\frac{1}{(s-a)^n} \right] = \frac{e^{at} t^{n-1}}{(n-1)!}.$$

$$(5) L^{-1} \left(\frac{1}{s^2 + a^2} \right) = \frac{1}{a} \sin at.$$

$$(6) L^{-1} \left(\frac{s}{s^2 + a^2} \right) = \cos at.$$

$$(7) L^{-1} \left(\frac{1}{s^2 - a^2} \right) = \frac{1}{a} \sinh at.$$

$$(8) L^{-1} \left(\frac{s}{s^2 - a^2} \right) = \cosh at.$$

$$(9) L^{-1} \left[\frac{1}{(s-a)^2 + b^2} \right] = \frac{1}{b} e^{at} \sin bt.$$

$$(10) L^{-1} \left[\frac{s-a}{(s-a)^2 + b^2} \right] = e^{at} \cos bt.$$

$$(11) L^{-1} \left(\frac{s}{(s^2 + a^2)^2} \right) = \frac{1}{2a} t \sin at.$$

$$(12) L^{-1} \left[\frac{1}{(s^2 + a^2)^2} \right] = \frac{1}{2a^3} (\sin at - at \cos at).$$

The reader is strongly advised to commit these results to memory. The results (1) to (10) follow at once from their corresponding results in § 21.3 and 21.4. As illustrations, we shall prove (11) and (12). Example 21.4 gives

$$L(t \sin at) = \frac{2as}{(s^2 + a^2)^2} \text{ and } L(t \cos at) = \frac{s^2 - a^2}{(s^2 + a^2)^2}$$

$$\therefore t \sin at = 2a L^{-1} \left[\frac{s}{(s^2 + a^2)^2} \right], \text{ whence follows (11).}$$

$$\begin{aligned} \text{Also } t \cos at &= L^{-1} \left[\frac{s^2 - a^2}{(s^2 + a^2)^2} \right] = L^{-1} \left[\frac{(s^2 + a^2) - 2a^2}{(s^2 + a^2)^2} \right] \\ &= L^{-1} \left[\frac{1}{s^2 + a^2} \right] - 2a^2 L^{-1} \left[\frac{1}{(s^2 + a^2)^2} \right] \\ &= \frac{1}{a} \sin at - 2a^2 L^{-1} \left[\frac{1}{(s^2 + a^2)^2} \right] \text{ whence follows (12).} \end{aligned}$$

Obs. Go through the note on the 'partial fractions' given in para 10 of 'useful information' in Appendix I.

Example 21.17. Find the inverse transforms of

$$(i) \frac{s^2 - 3s + 4}{s^3}$$

$$(ii) \frac{s+2}{s^2 - 4s + 13}$$

(V.T.U., 2008)

$$\text{Solution. (i) } L^{-1} \left(\frac{s^2 - 3s + 4}{s^3} \right) = L^{-1} \left(\frac{1}{s} \right) - 3L^{-1} \left(\frac{1}{s^2} \right) + 4L^{-1} \left(\frac{1}{s^3} \right) = 1 - 3t + 4 \cdot t^2/2! = 1 - 3t + 2t^2.$$

$$\begin{aligned} (ii) \quad L^{-1} \left(\frac{s+2}{s^2 - 4s + 13} \right) &= L^{-1} \left[\frac{s+2}{(s-2)^2 + 9} \right] = L^{-1} \left[\frac{s-2+4}{(s-2)^2 + 3^2} \right] \\ &= L^{-1} \left[\frac{s-2}{(s-2)^2 + 3^2} \right] + 4L^{-1} \left[\frac{1}{(s-2)^2 + 3^2} \right] = e^{2t} \cos 3t + \frac{4}{3} e^{2t} \sin 3t. \end{aligned}$$

Example 21.18. Find the inverse transforms of

$$(i) \frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6}$$

(V.T.U., 2007 ; U.P.T.U., 2004)

$$(ii) \frac{4s+5}{(s-1)^2(s+2)}$$

(Kurukshetra, 2005)

Solution. (i) Here the denominator = $(s - 1)(s - 2)(s - 3)$.

So let
$$\frac{2s^2 - 6s + 5}{(s - 1)(s - 2)(s - 3)} = \frac{A}{s - 1} + \frac{B}{s - 2} + \frac{C}{s - 3}$$

Then $A = [2 \cdot 1^2 - 6 \cdot 1 + 5]/(1 - 2)(1 - 3) = \frac{1}{2}$

$B = [2 \cdot 2^2 - 6 \cdot 2 + 5]/(2 - 1)(2 - 3) = -1$

and $C = [2 \cdot 3^2 - 6 \cdot 3 + 5]/(3 - 1)(3 - 2) = \frac{5}{2}$.

$\therefore L^{-1} \left(\frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6} \right) = \frac{1}{2} L^{-1} \left(\frac{1}{s - 1} \right) - L^{-1} \left(\frac{1}{s - 2} \right) + \frac{5}{2} L^{-1} \left(\frac{1}{s - 3} \right)$
 $= \frac{1}{2} e^t - e^{2t} + \frac{5}{2} e^{3t}.$

(ii) Let
$$\frac{4s + 5}{(s - 1)^2 (s + 2)} = \frac{A}{s - 1} + \frac{B}{(s - 1)^2} + \frac{4(-2) + 5}{(-2 - 1)^2 (s + 2)}$$

Multiplying both sides by $(s - 1)^2 (s + 2)$, $4s + 5 = A(s - 1)(s + 2) + B(s + 2) - \frac{1}{3}(s - 1)^2$

Putting $s = 1$, $9 = 3B$, $\therefore B = 3$.

Equating the coefficients of s^2 from both sides,

$0 = A - \frac{1}{3}$, $\therefore A = \frac{1}{3}$.

$\therefore L^{-1} \left[\frac{4s + 5}{(s - 1)^2 (s + 2)} \right] = \frac{1}{3} L^{-1} \left(\frac{1}{s - 1} \right) + 3 L^{-1} \left[\frac{1}{(s - 1)^2} \right] - \frac{1}{3} L^{-1} \left(\frac{1}{s + 2} \right)$
 $= \frac{1}{3} e^t + 3te^t - \frac{1}{3} e^{-2t}.$

Example 21.19. Find the inverse transforms of

(i) $\frac{5s + 3}{(s - 1)(s^2 + 2s + 5)}$

(Rohtak, 2009 ; U.P.T.U., 2005)

(ii) $\frac{s}{s^4 + 4a^4}$

(Mumbai, 2008)

Solution. (i) Let
$$\frac{5s + 3}{(s - 1)(s^2 + 2s + 5)} = \frac{5(1) + 3}{(s - 1)(1^2 + 2 \cdot 1 + 5)} + \frac{As + B}{s^2 + 2s + 5}$$

Multiplying both sides by $(s - 1)(s^2 + 2s + 5)$,

$5s + 3 = 1 \cdot (s^2 + 2s + 5) + (As + B)(s - 1).$

Equating the coefficients of s^2 from both sides,

$0 = 1 + A$, $\therefore A = -1$.

Putting $s = 0$, $3 = 5 - B$, $\therefore B = 2$.

$\therefore L^{-1} \left[\frac{5s + 3}{(s - 1)(s^2 + 2s + 5)} \right] = L^{-1} \left(\frac{1}{s - 1} \right) + L^{-1} \left(\frac{-s + 2}{s^2 + 2s + 5} \right)$
 $= L^{-1} \left(\frac{1}{s - 1} \right) + L^{-1} \left[\frac{-(s + 1) + 3}{(s + 1)^2 + 4} \right] = L^{-1} \left(\frac{1}{s - 1} \right) - L^{-1} \left[\frac{s + 1}{(s + 1)^2 + 2^2} \right] + 3 L^{-1} \left[\frac{1}{(s + 1)^2 + 2^2} \right]$
 $= e^t - e^{-t} \cos 2t + \frac{3}{2} e^{-t} \sin 2t.$

(ii) Since $s^4 + 4a^4 = (s^2 + 2a^2)^2 - (2as)^2 = (s^2 + 2as + 2a^2)(s^2 - 2as + 2a^2)$

$\therefore$ Let
$$\frac{s}{s^4 + 4a^4} = \frac{As + B}{s^2 + 2as + 2a^2} + \frac{Cs + D}{s^2 - 2as + 2a^2}$$

Multiplying both sides by $s^4 + 4a^4$,

$s = (As + B)(s^2 - 2as + 2a^2) + (Cs + D)(s^2 + 2as + 2a^2)$

Equating coefficients of s^3 , $0 = A + C$... (i)

Equating coefficients of s^2 , $0 = -2aA + B + 2aC + D$... (ii)

Equating coefficients of s , $1 = 2a^2A - 2aB + 2a^2C + 2aD$... (iii)

Putting $s = 0$, $0 = 2a^2B + 2a^2D$... (iv)

From (iv), $B + D = 0$... (v)

$\therefore$ (ii) becomes $-A + C = 0$, and by (i), we get $A = C = 0$.

Then (iii) reduces to $D - B = 1/2a$ and by (v), $B = -1/4a$, $D = 1/4a$.

$$\begin{aligned}\therefore L^{-1}\left(\frac{s}{s^4 + 4a^4}\right) &= -\frac{1}{4a} L^{-1}\left(\frac{1}{s^2 + 2as + 2a^2}\right) + \frac{1}{4a} L^{-1}\left(\frac{1}{s^2 - 2as + 2a^2}\right) \\ &= -\frac{1}{4a} L^{-1}\left[\frac{1}{(s+a)^2 + a^2}\right] + \frac{1}{4a} L^{-1}\left[\frac{1}{(s-a)^2 + a^2}\right] \\ &= -\frac{1}{4a} \cdot \frac{1}{a} e^{-at} \sin at + \frac{1}{4a} \cdot \frac{1}{a} e^{at} \sin at = \frac{1}{2a^2} \sin at \left(\frac{e^{at} - e^{-at}}{2}\right) = \frac{1}{2a^2} \sin at \sinh at.\end{aligned}$$

PROBLEMS 21.4

Find the inverse Laplace transforms of :

- $\frac{2s-5}{4s^2+25} + \frac{4s-18}{9-s^2}$ (S.V.T.U., 2008)
- $\frac{s}{(2s-1)(3s-1)}$ (V.T.U., 2010)
- $\frac{3s+2}{s^2-s-2}$ (V.T.U., 2010 S)
- $\frac{1-7s}{(s-3)(s-1)(s+2)}$ (B.P.T.U., 2005 S)
- $\frac{2p^2-6p+5}{p^3-6p^2+11p-6}$ (U.P.T.U., 2004)
- $\frac{s^2-10s+13}{(s-7)(s^2-5s+6)}$
- $\frac{s}{(s^2-1)^2}$ (Kurukshetra, 2005)
- $\frac{1+2s}{(s+2)^2(s-1)^2}$
- $\frac{s^3}{s^4-a^4}$ (Kurukshetra, 2005)
- $\frac{1}{s^3-a^3}$
- $\frac{s^2+6}{(s^2+1)(s^2+4)}$
- $\frac{2s-3}{s^2+4s+13}$
- $\frac{s^2+s}{(s^2+1)(s^2+2s+2)}$
- $\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}$ (Mumbai, 2008)
- $\frac{s}{s^4+s^2+1}$ (Raipur, 2005)
- $\frac{a(s^2-2a^2)}{s^4+4a^4}$ (Mumbai, 2009)

21.13 OTHER METHODS OF FINDING INVERSE TRANSFORMS

We have seen that the most effective method of finding the inverse transforms is by means of partial fractions. However, various other methods are available which depend on the following *important inversion formulae*.

I. Shifting property for inverse Laplace transforms.

If $L^{-1}[\bar{f}(s)] = f(t)$, then

$$L^{-1}[\bar{f}'(s - a)] = e^{at} f(t) = e^{at} L^{-1}[\bar{f}(s)].$$

II. If $L^{-1}[\bar{f}(s)] = f(t)$ and $f(0) = 0$, then

$$L^{-1}[s \bar{f}(s)] = \frac{d}{dt} \{f(t)\}$$

In general, $L^{-1}[s^n \bar{f}(s)] = \frac{d^n}{dt^n} \{f(t)\}$ provided $f(0) = f'(0) = \dots = f^{n-1}(0) = 0$.

The above formulae at once follow from the results of § 21.7 (Transforms of derivatives).

III. If $L^{-1}[\bar{f}(s)] = f(t)$, then

$$L^{-1}\left\{\frac{\bar{f}(s)}{s}\right\} = \int_0^t f(t) dt$$

This result follows from § 21.8 (Transforms of integrals)

Also
$$L^{-1}\left\{\frac{\bar{f}(s)}{s^2}\right\} = \int_0^t \left\{\int_0^t f(t) dt\right\} dt$$

$$L^{-1}\left\{\frac{\bar{f}(s)}{s^3}\right\} = \int_0^t \left\{\int_0^t \left(\int_0^t f(t) dt\right) dt\right\} dt \text{ and so on.}$$

IV. If $L^{-1}[\bar{f}(s)] = f(t)$, then

$$t f(t) = L^{-1}\left\{-\frac{d}{ds} [\bar{f}(s)]\right\}$$

This result follows from $L\{t f(t)\} = -\frac{d}{ds} \{\bar{f}(s)\}$ (§ 21.9)

V. The formula of § 21.10, i.e.,

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty \bar{f}(s) ds$$

is useful in finding $f(t)$ when $f(s)$ is given, provided the inverse transform of $\int_s^\infty \bar{f}(s) ds$ can be conveniently calculated.

Example 21.20. Find the inverse Laplace transforms of the following :

(i) $\frac{s^2}{(s-2)^3}$

(ii) $\frac{s+3}{s^2-4s+13}$

(iii) $\frac{(s+2)^2}{(s^2+4s+8)^2}$

(Mumbai, 2005)

Solution. (i) Since $s^2 = (s-2)^2 + 4(s-2) + 4$

$$\therefore \frac{s^2}{(s-2)^3} = \frac{1}{s-2} + \frac{4}{(s-2)^2} + \frac{4}{(s-2)^3}$$

$$\therefore L^{-1}\left\{\frac{s^2}{(s-2)^3}\right\} = L^{-1}\left\{\frac{1}{s-2}\right\} + 4L^{-1}\left\{\frac{1}{(s-2)^2}\right\} + 4L^{-1}\left\{\frac{1}{(s-2)^3}\right\}$$

$$= e^{2t} + 4e^{2t}t + 2e^{2t}t^2.$$

[using shifting property]

(ii)
$$\frac{s+3}{s^2-4s+13} = \frac{s-2}{(s-2)^2+3^2} + \frac{5}{(s-2)^2+3^2}$$

$$\therefore L^{-1}\left\{\frac{s+3}{s^2-4s+13}\right\} = L^{-1}\left\{\frac{s-2}{(s-2)^2+3^2}\right\} + \frac{5}{3}L^{-1}\left\{\frac{3}{(s-2)^2+3^2}\right\}$$

$$= e^{2t} \cos 3t + \frac{5}{3} e^{2t} \sin 3t.$$

[Using shifting property]

$$\begin{aligned} \text{(iii)} \quad L^{-1} \frac{(s+2)^2}{(s^2+4s+8)^2} &= L^{-1} \frac{(s+2)^2}{(s^2+4s+4+4)^2} = L^{-1} \frac{(s+2)^2}{[(s+2)^2+4]^2} \\ &= e^{-2t} L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\} = e^{-2t} L^{-1} \left\{ \frac{s^2+4-4}{(s^2+4)^2} \right\} \\ &= e^{-2t} L^{-1} \left\{ \frac{1}{s^2+4} - \frac{4}{(s^2+4)^2} \right\} = \frac{e^{-2t} \sin 2t}{2} - 4e^{-2t} L^{-1} \left\{ \frac{1}{(s^2+4)^2} \right\} \\ &= \frac{e^{-2t} \sin 2t}{2} - 4e^{-2t} \left\{ \frac{1}{4} \left(\frac{\sin 2t}{4} - \frac{t \cos 2t}{2} \right) \right\} \\ &= e^{-2t} \left\{ \frac{\sin 2t}{2} - \frac{\sin 2t}{4} + \frac{t \cos 2t}{2} \right\} = e^{-2t} \left\{ \frac{\sin 2t}{4} + \frac{t \cos 2t}{2} \right\}. \end{aligned}$$

Example 21.21. Find the inverse transform of (i) $1/s(s^2+a^2)$
(ii) $1/s(s+a)^3$.

(P.T.U., 2003)

Solution. (i) Since $L^{-1} \left(\frac{1}{s^2+a^2} \right) = \frac{1}{a} \sin at$.

therefore, by formula III above,

$$L^{-1} \left\{ \frac{1}{s(s^2+a^2)} \right\} = \int_0^t \frac{1}{a} \sin at \, dt = \frac{1}{a^2} [-\cos at]_0^t = (1 - \cos at)/a^2$$

$$\text{(ii)} \quad L^{-1} \left\{ \frac{1}{s(s+a)^3} \right\} = L^{-1} \left\{ \frac{1}{[(s+a)-a](s+a)^3} \right\} = e^{-at} L^{-1} \left\{ \frac{1}{(s-a)s^3} \right\}$$

$$\text{Now} \quad L^{-1} \left\{ \frac{1}{s-a} \right\} = e^{at} \quad \therefore \quad L^{-1} \left\{ \frac{1}{(s-a)s} \right\} = \int_0^t e^{at} \, dt = \frac{e^{at}}{a} - \frac{1}{a}, \text{ by III above}$$

$$\therefore \quad L^{-1} \left\{ \frac{1}{(s-a)s^2} \right\} = \frac{1}{a} \int_0^t (e^{at} - 1) \, dt = \frac{1}{a^2} (e^{at} - at - 1)$$

$$L^{-1} \left\{ \frac{1}{(s-a)s^3} \right\} = \frac{1}{a^2} \int_0^t (e^{at} - at - 1) \, dt = \frac{1}{a^3} \left(e^{at} - \frac{a^2}{2} t^2 - at - 1 \right)$$

$$\text{Hence} \quad L^{-1} \left\{ \frac{1}{s(s+a)^3} \right\} = e^{-at} \cdot \frac{1}{a^3} \left(e^{at} - \frac{a^2 t^2}{2} - at - 1 \right) = \frac{1}{a^3} \left(1 - e^{-at} - ate^{-at} - \frac{a^2}{2} t^2 e^{-at} \right).$$

Example 21.22. Find the inverse Laplace transforms of:

$$\text{(i)} \quad \frac{s}{(s^2+a^2)^2} \quad (\text{S.V.T.U., 2009}) \quad \text{(ii)} \quad \frac{s^2}{(s^2+a^2)^2} \quad (\text{Hazaribag, 2009}) \quad \text{(iii)} \quad \frac{1}{(s^2+a^2)^2}.$$

Solution. (i) If $f(t) = L^{-1} \frac{s}{(s^2+a^2)^2}$, then by formula V above,

$$L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty \frac{s}{(s^2+a^2)^2} \, ds = \frac{1}{2} \int_s^\infty \frac{2s}{(s^2+a^2)^2} \, ds = -\frac{1}{2} \left(\frac{1}{s^2+a^2} \right)_s^\infty = \frac{1}{2} \cdot \frac{1}{s^2+a^2}$$

$$\therefore \quad \frac{f(t)}{t} = \frac{1}{2} L^{-1} \left(\frac{1}{s^2+a^2} \right) = \frac{\sin at}{2a}$$

Hence, $f(t) = \frac{1}{2a} t \sin at$.

Otherwise : Let $f(t) = L^{-1} \left\{ \frac{1}{s^2 + a^2} \right\} = \frac{\sin at}{a}$ so that $\bar{f}(s) = \frac{1}{s^2 + a^2}$

Then by (IV) above, $t f(t) = L^{-1} \left\{ -\frac{d}{ds} [\bar{f}(s)] \right\} = L^{-1} \left\{ -\frac{d}{ds} \left(\frac{1}{s^2 + a^2} \right) \right\}$

or

$$\frac{t \sin at}{a} = L^{-1} \left\{ \frac{2s}{(s^2 + a^2)^2} \right\}. \text{ Hence } L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\} = \frac{1}{2a} t \sin at.$$

(ii) In (i), we have proved that

$$L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\} = \frac{1}{2a} t \sin at = f(t), \text{ say}$$

Since $f(0) = 0$, we get from formula II above, that

$$\begin{aligned} L^{-1} \left\{ \frac{s^2}{(s^2 + a^2)^2} \right\} &= L^{-1} \left\{ s \cdot \frac{s}{(s^2 + a^2)^2} \right\} = \frac{d}{dt} \{f(t)\} \\ &= \frac{d}{dt} \left(\frac{1}{2a} t \sin at \right) = \frac{1}{2a} (\sin at + at \cos at) \end{aligned}$$

(iii) In (i), we have shown that

$$L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\} = \frac{1}{2a} (t \sin at) = f(t), \text{ say}$$

By formula III above, we have

$$\begin{aligned} L^{-1} \left\{ \frac{1}{(s^2 + a^2)^2} \right\} &= L^{-1} \left\{ \frac{1}{s} \cdot \frac{s}{(s^2 + a^2)^2} \right\} = \int_0^t f(t) dt = \int_0^t \frac{t \sin at}{2a} dt \\ &= \frac{1}{2a} \left\{ t \cdot \frac{-\cos at}{a} \Big|_0^t - \int_0^t 1 \cdot \left(\frac{-\cos at}{a} \right) dt \right\} \\ &= \frac{1}{2a} \left\{ \frac{-t \cos at}{a} + \frac{\sin at}{a^2} \right\} = \frac{1}{2a^3} (\sin at - at \cos at). \end{aligned}$$

Example 21.23. Find the inverse Laplace transforms of

$$(i) \frac{s+2}{s^2(s+1)(s-2)} \quad (V.T.U., 2003) \quad (ii) \frac{s+2}{(s^2+4s+5)^2} \quad (S.V.T.U., 2009; P.T.U., 2005)$$

Solution. (i) $L^{-1} \left\{ \frac{s+2}{(s+1)(s-2)} \right\} = \frac{4}{3} L^{-1} \left(\frac{1}{s-2} \right) - \frac{1}{3} L^{-1} \left(\frac{1}{s+1} \right) = \frac{4}{3} e^{2t} - \frac{1}{3} e^{-t}$

By III above, $L^{-1} \left\{ \frac{s+2}{s(s+1)(s-2)} \right\} = \int_0^t L^{-1} \left(\frac{s+2}{(s+1)(s-2)} \right) dt$

$$= \int_0^t \left(\frac{4}{3} e^{2t} - \frac{1}{3} e^{-t} \right) dt = \frac{2}{3} e^{2t} + \frac{1}{3} e^{-t} - 1$$

Again by III above, $L^{-1} \frac{s+2}{s^2(s+1)(s-2)} = \int_0^t L^{-1} \left\{ \frac{s+2}{s(s+1)(s-2)} \right\} dt$

$$= \int_0^t \left(\frac{2}{3} e^{2t} + \frac{1}{3} e^{-t} - 1 \right) dt = \frac{1}{3} (e^{2t} - e^{-t} - t).$$

$$(ii) \quad L^{-1} \left(\frac{1}{s^2 + 4s + 5} \right) = L^{-1} \left\{ \frac{1}{(s+2)^2 + 1} \right\} = e^{-2t} \sin t$$

$$\text{By II above, } L^{-1} \left\{ \frac{d}{ds} \left(\frac{1}{s^2 + 4s + 5} \right) \right\} = (-1)^1 t \cdot e^{-2t} \sin t$$

$$\text{i.e.,} \quad L^{-1} \left\{ \frac{-(2s+4)}{(s^2 + 4s + 5)^2} \right\} = -t \cdot e^{-2t} \sin t$$

$$\text{or} \quad L^{-1} \left\{ \frac{s+2}{(s^2 + 4s + 5)^2} \right\} = \frac{1}{2} t \cdot e^{-2t} \sin t.$$

Example 21.24. Find the inverse Laplace transforms of the following :

$$(i) \log \frac{s+1}{s-1} \quad (\text{S.V.T.U., 2009 ; Bhopal, 2008}) \quad (ii) \log \frac{s^2+1}{s(s+1)} \quad (\text{S.V.T.U., 2009 ; V.T.U., 2008})$$

$$(iii) \cot^{-1} \left(\frac{s}{2} \right) \quad (iv) \tan^{-1} \left(\frac{2}{s^2} \right). \quad (\text{V.T.U., 2011 ; Mumbai, 2005 S})$$

Solution. (i) If $f(t) = L^{-1} \log \frac{s+1}{s-1}$, then by IV above,

$$\begin{aligned} t f(t) &= L^{-1} \left\{ -\frac{d}{ds} \log \left(\frac{s+1}{s-1} \right) \right\} = -L^{-1} \left\{ \frac{d}{ds} \log (s+1) \right\} + L^{-1} \left\{ \frac{d}{ds} \log (s-1) \right\} \\ &= -L^{-1} \left(\frac{1}{s+1} \right) + L^{-1} \left(\frac{1}{s-1} \right) = -e^{-t} + e^t = 2 \sinh t \end{aligned}$$

Thus $f(t) = (2 \sinh t)/t$.

(ii) If $f(t) = L^{-1} \log \frac{s^2+1}{s(s+1)}$, then by IV above,

$$\begin{aligned} t f(t) &= L^{-1} \left\{ -\frac{d}{ds} \log \left(\frac{s^2+1}{s(s+1)} \right) \right\} = -L^{-1} \left\{ \frac{d}{ds} \log (s^2+1) \right\} + L^{-1} \left\{ \frac{d}{ds} \log s \right\} \\ &\quad + L^{-1} \left\{ \frac{d}{ds} \log (s+1) \right\} \\ &= -L^{-1} \left(\frac{2s}{s^2+1} \right) + L^{-1} \left(\frac{1}{s} \right) + L^{-1} \left(\frac{1}{s+1} \right) = -2 \cos t + 1 + e^{-t} \end{aligned}$$

Thus $f(t) = \frac{1}{t} (1 + e^{-t} - 2 \cos t)$.

(iii) If $f(t) = L^{-1} \cot^{-1} \left(\frac{s}{2} \right)$, then by IV above,

$$t f(t) = L^{-1} \left\{ -\frac{d}{ds} \cot^{-1} \left(\frac{s}{2} \right) \right\} = L^{-1} \left(\frac{2}{s^2 + 2^2} \right) = \sin 2t$$

Thus $f(t) = (\sin 2t)/t$.

(iv) If $f(t) = L^{-1} \left(\tan^{-1} \frac{2}{s^2} \right)$, then by IV above,

$$t f(t) = L^{-1} \left\{ -\frac{d}{ds} \tan^{-1} \left(\frac{2}{s^2} \right) \right\} = L^{-1} \left\{ \frac{4s}{s^4 + 4} \right\}$$

$$\begin{aligned}
 &= L^{-1} \left\{ \frac{4s}{(s^2 + 2)^2 - (2s)^2} \right\} = L^{-1} \left\{ \frac{4s}{(s^2 + 2 + 2s)(s^2 + 2 - 2s)} \right\} \\
 &= L^{-1} \left\{ \frac{1}{s^2 - 2s + 2} - \frac{1}{s^2 + 2s + 2} \right\} = L^{-1} \left\{ \frac{1}{(s-1)^2 + 1} - \frac{1}{(s+1)^2 + 1} \right\} \\
 &= e^t \sin t - e^{-t} \sin t = 2 \sinh t \sin t.
 \end{aligned}$$

21.14 CONVOLUTION THEOREM

If $L^{-1} \{ \bar{f}(s) \} = f(t)$, and $L^{-1} \{ \bar{g}(s) \} = g(t)$,

then $L^{-1} \{ \bar{f}(s) \bar{g}(s) \} = \int_0^t f(u) g(t-u) du = F * G$

[$F * G$ is called the **convolution** or **fasting** of F and G .]

Let $\phi(t) = \int_0^t f(u) g(t-u) du$

$$L\{\phi(t)\} = \int_0^\infty e^{-st} \left\{ \int_0^t f(u) g(t-u) du \right\} dt = \int_0^\infty \int_0^t e^{-st} f(u) g(t-u) du dt \quad \dots(1)$$

The domain of integration for this double integral is the entire area lying between the lines $u = 0$ and $u = t$ (Fig. 21.2).

On changing the order of integration, we get

$$\begin{aligned}
 L\{\phi(t)\} &= \int_0^\infty \int_u^\infty e^{-st} f(u) g(t-u) dt du \\
 &= \int_0^\infty e^{-su} f(u) \left\{ \int_0^\infty e^{-s(t-u)} g(t-u) dt \right\} du \\
 &= \int_0^\infty e^{-su} f(u) \left\{ \int_0^\infty e^{-sv} g(v) dv \right\} du \text{ on putting } t-u=v \\
 &= \int_0^\infty e^{-su} f(u) g(s) du = \int_0^\infty e^{-su} f(u) du \cdot \bar{g}(s) \\
 &= \bar{f}(s) \cdot \bar{g}(s) \text{ whence follows the desired result.}
 \end{aligned}$$

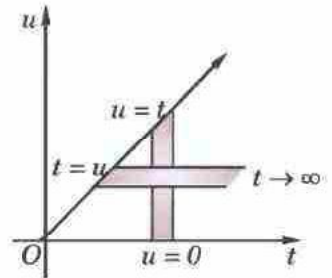


Fig. 21.2

Example 21.25. Apply Convolution theorem to evaluate

(i) $L^{-1} \frac{s}{(s^2 + a^2)^2}$

(V.T.U., 2010)

(ii) $L^{-1} \frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$

(V.T.U., 2011 S ; Bhopal, 2008 ; Mumbai, 2007)

Solution. (i) Since $f(t) = L^{-1} \left(\frac{s}{s^2 + a^2} \right) = \cos at$ and $g(t) = L^{-1} \left(\frac{s}{s^2 + a^2} \right) = \frac{1}{a} \sin at$

$\therefore$ by Convolution theorem, we get

$$\begin{aligned}
 L^{-1} \left[\frac{s}{s^2 + a^2} \cdot \frac{1}{s^2 + a^2} \right] &= \int_0^t \cos au \frac{\sin a(t-u)}{a} du \quad \left[\begin{array}{l} \because f(u) = \cos au \\ g(t-u) = \frac{1}{a} \sin a(t-u) \end{array} \right] \\
 &= \frac{1}{2a} \int_0^t [\sin at - \sin (2au - at)] dt = \frac{1}{2a} \left[u \sin at + \frac{1}{2a} \cos (2au - at) \right]_0^t = \frac{1}{2a} t \sin at
 \end{aligned}$$

Hence $L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\} = \frac{1}{2a} t \sin at.$